**Table 19.** Reference architecture components and candidate technologies

| ID | Component | Function | Candidate open technologies | Reference implementation |
|---|---|---|---|---|
| C1 | Source connectors | Harvest public documents on a schedule, respecting site terms; record provenance | Python workers, Scrapy, cron or Airflow | Planned |
| C2 | Ingestion and normalisation | Extract text from PDF and HTML, Bangla OCR, Unicode normalisation, chunking | Apache Tika [62], Tesseract (ben, eng) [63], custom normaliser | bapil.text, bapil.corpus |
| C3 | Document store and catalogue | Originals, extracted text and DCAT-aligned metadata with versioning | PostgreSQL [64], MinIO (S3-compatible), CKAN [65] | JSONL corpus with checksums |
| C4 | Search index | Lexical BM25 plus dense vectors with hybrid ranking | OpenSearch [66], or PostgreSQL with pgvector [67] | bapil.search (BM25, bilingual) |
| C5 | Answer service | Retrieve, cite, abstain; optional LLM generation with citation verification | Open-weight LLMs served with vLLM or llama.cpp | bapil.answer (extractive, cited) |
| C6 | Analytics engine | Budget extraction, indicators, topic models | Python, pandas, DuckDB | Planned |
| C7 | Portal and dashboards | Bilingual, accessible user interface | Web framework; Apache Superset | Planned |
| C8 | API gateway | Versioned REST API, quotas, rate limits | FastAPI, Kong | bapil.api (standard library) |
| C9 | Identity and access | Single sign-on, role-based access, MFA | Keycloak | Planned |
| C10 | Audit and observability | Tamper-evident logs, metrics, tracing | OpenTelemetry, Prometheus, Grafana | bapil.audit (hash-chained JSONL) |
| C11 | Model registry and evaluation | Model cards, versioned evaluations, benchmarks | MLflow | bapil.evaluate |
| C12 | Governance workbench | AI system register, risk assessments, incidents, transparency register | Lightweight web application | bapil.risk, bapil.registry |

*Technologies are candidates, not mandates; final choices are made by the Technical Architecture Board against requirements NFR-02, NFR-03 and NFR-06.*

Source: BAPIL Technical Blueprint, Version 1.0 (2026). CC BY 4.0.